ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) Assumptions: pinhole (thin-lens) camera model, lens focused on the belt plane, belt plane perpendicular to the optical axis, no lens distortion, so every pixel images the same length on the belt.

(i) The focal length in pixel units is the physical focal length divided by the pixel pitch: $f_x = f/p = (16\times10^{-3}\text{ m})/(5\times10^{-6}\text{ m}) = 3200$. The metres cancel so the answer is in pixels. **f_x = 3200 px**.

(ii) Width of the sensor = N·p = 1280 × 5 µm = 6400 µm = 6.4 mm. Using similar triangles between the sensor and the belt plane, $W_s/f = FOV/Z$, hence $FOV = W_s \cdot Z/f = (6.4\text{ mm})(400\text{ mm})/(16\text{ mm}) = 160$ mm. **FOV = 160 mm**.

Ground sampling distance = FOV/N = 160 mm / 1280 px = 0.125 mm/px. Cross-check with the pixel-based formula Z/f_x = 400 mm / 3200 px = 0.125 mm/px; both agree, so **1 pixel = 0.125 mm = 125 µm** on the belt.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S$ = 1.2207 mV / (10 mV/N) = 0.12207 N. **$\Delta F_{min} \approx 0.122$ N**. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) With u held constant over [kT, (k+1)T] (zero-order hold) the variation-of-constants formula gives $x((k+1)T) = e^{AT}x(kT) + \int_0^T e^{A(T-s)}B\,u(kT)\,ds$. Since u(kT) is constant it leaves the integral, so $F = e^{AT}$ and $G = \int_0^T e^{A\sigma}B\,d\sigma = A^{-1}(e^{AT} - I)B$ for invertible A.

Substituting A = −2, B = 1, T = 0.1 s: $F = e^{-0.2} = 0.81873$. $G = (0.81873 - 1)/(-2) = 0.090635$.

Sanity check: the DC gain of the continuous plant is $-A^{-1}B = 0.5$, and the discrete DC gain $G/(1 - F) = 0.090635/0.18127 = 0.5$ ✓.

$x_{k+1} = 0.8187\,x_k + 0.0906\,u_k$

Question 3

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is $T = 1/f_s$ = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\text{ s}^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a) z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a) z^{-1}]$.

Dividing through by $(1 + a)$: $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}}$. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant ($z = 1$): $y = 2(0.05912)u + 0.88177\,y \Rightarrow y(1 - 0.88177) = 0.11824\,u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

(c) First check observability: $O = [C; CA] = [1, 0; 0, 1]$, full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and $C = [1, 0]$, $LC = [\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\,s + (2\ell_1 + \ell_2)$.

Both estimator pols at $s = -10$ means the desired polynomial is $(s + 10)^2 = s^2 + 20\,s + 100$. Comparing coefficients: s^1: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. s^0: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

$\mathbf{L = [18, 64]^T}$. Verification: $A - LC = [-18, 1; -64, -2]$ has trace -20 and determinant $(-18)(-2) - (1)(-64) = 36 + 64 = 100$, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

(d) Observer advantage: because it is a copy of the plant corrected by the encoder error, it gives a velocity estimate without the delay that a filter needs for noice rejection, and it can also estimate quantities no filter on θ can, such as an unknown load torque.

A filter is the safer choice when the model is wrong or the true input is not known, e.g. unmodelled friction or backlash, a changing payload, or actuator saturation: the observer would then be driven by a wrong u and drift, while the filter only uses measured data.

Sampling at 1 kHz is fast compared with the 10 rad/s observer bandwidth, so discretisation effects are small.

Question 4

Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\,u$ with $a = 2$, $c = 6$, $b = 1$; controller $u = N_{ff}\,r - k_1\theta - k_2\dot{\theta}$.

(a) Closed loop: $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $s^2 + 4s + 8 = 0$.

$s = [-4 \pm \sqrt{(16 - 32)}]/2 = -4/2 \pm \sqrt{(-16)} = \mathbf{-2 \pm 4j}$.

DC gain $= N_{ff}/8 = 1$, so $\mathbf{N_{ff} = 8}$.

(b) With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

(c) Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(Result)**

Desired poles -4 and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: s^2: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. s^1: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. s^0: $b K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

(d) Suppose the closed loop is stabel and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value ξ_{ss}). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\mathbf{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of ξ_{ss} needed to hold the equilibrum (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward N_{ff} scheme of part (b)).

Practical failure mode: integrater windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the actuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrator.

Note: the integrator adds one state, so the loop becomes third order.